Weather Or Not - Mobile App Development 2025/2026

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App Description:
Our app helps users choose the perfect outfit by suggesting clothing combinations from their personal wardrobe based on the current weather at their current location or a selected location. This saves valuable time in the morning and removes the stress of deciding what to wear. Users can set individual temperature preferences, defining at which temperatures they feel cold or warm. Wardrobe items can be added with pre-defined categories and parameters, which can also be adjusted for the algorithm. In addition, activities can be planned in a monthly calendar, which also influences the outfit suggestions to ensure they are suitable for the day's plans.

Emil:
- Initial design of screens and layouts
- Research and connection with DWD API, fetching and storing weather data
- Weather overview screen and weather card widget
- Location picker implementation, including asking for the user's location and finding the next best fallback stations, if there is no weather data for a selected location
- Recurring activities feature
- Bug fixing and minor adjustments at various points

Till:
- Outfit suggestion algorithm using a point-based scoring system including weather constraints (temperature, wind, rain), activity constraints (activities & daily commuting), formality, and style constraints (style tags, color) (and a little bit of randomness).
- Reroll/refresh feature for outfit suggestions
- wardrobe categories, subcategories, and attributes
- Bug fixing

Ahsan:
- Initial project setup, including the creation of boilerplate screens: Planner, Home, Wardrobe, and Settings. Hooked everything into a functional navigation bar.
- Built a complete wardrobe system with a searching feature where users can add, edit, and delete clothes.
- Added a favorite tagging system that automatically surfaces favorited items in a separate screen.
- Built the save-outfit feature so users can store generated outfits and view them later.
- Added the InWash status so users can temporarily flag items.
- Show the total number of items in the wardrobe.

Maja:

- Home screen with weather display and BrightSky API integration
  (The API was later changed to the DWD API by Emil to better meet the requirements)
- Daily activities feature (activities were later updated and refined by Till)
- Settings page and functionality
- Monthly planner with activities and synchronization with the home screen
- Logo design and implementation

Challenges We Faced (and fixed):

- Not every DWD weather station is returning (complete) data. Therefore, we added some fallback mechanisms, including fallbacks to the next nearest stations within a certain radius, if a station doesn't have data, and interpolating "live" weather data based on forecasts.
- We had synchronization issues between the monthly planner and the home screen.
- The outfit algorithm repeatedly suggested the exact same outfits instead of varied ones for similar weather. Therefore, we added some randomness to the selection.
- Initially, the activities were not factored in correctly for the clothes selection, causing the activities to basically have no impact.
- The score-based algorithm suggested items like jackets & pullovers, even if it was too warm, and shorts, even if it was too cold. We tried to add some extra constraints to avoid that.
- Initially confusing UI in some parts. For example, we had too many unsorted activities and or categories in the selection dialogs, so we grouped them and added a search function.
- At some points, we had internal variable race conditions in the state management, causing the app to crash or not behave correctly.